Question

At room temperature, excess formaldehyde hydrate was added to a dichloromethane solution of $[(\text{n-C}_4\text{H}_9)_4\text{N}]_2\text{Mo}_2\text{O}_7$, and after sufficient reaction (solution 1), ethyl ether was added for crystallization; the crude product was purified by recrystallization to obtain compound $\mathbf{X}$. Further studies have shown that $\mathbf{X}$ is a salt compound, and its crystal contains only 1 kind of cation and 1 kind of anion; the formula weight of the anion is 638.8, and it has a multinuclear cluster structure; among them, all Mo atoms have the same chemical environment. In $\mathbf{X}$, the content (mass fraction) of each element is: C, 43.1%; H, 8.2%; N, 3.1%; O, 17.6%.

Which of the following statements are correct:

1. The number of H in the chemical formula of $\mathbf{X}$ does not exceed 110

2. Solution 1 is neutral, or weakly acidic/basic

3. In one anion of $\mathbf{X}$, there are 2 triply coordinated O

4. In molybdates formed by molybdenum, the number of terminal oxygen atoms connected to each molybdenum is usually no more than 2

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1,2

G. 1,3

H. 1,4

I. 2,3

J. 2,4

K. 3,4

L. 1,2,3

M. 1,2,4

N. 1,3,4

O. 2,3,4

P. 1,2,3,4

Answer

Correct Answer: **K**

Detailed Explanation

$\mathbf{X}$ contains only Mo in addition to C, H, N, and O;

The content of Mo is: $1 - 43.1\,\% - 8.2\,\% - 3.1\,\% - 17.6\,\% = 28.0\,\%$

The ratio of the number of N and Mo atoms in $\mathbf{X}$ is: $(3.1\,\%/14.01) : (28.0\,\%/95.95) = 0.758 \approx 3 : 4$

CHECKPOINT 0.5 PTS

$\mathrm{N} : \mathrm{Mo} = 3 : 4$ in $\mathbf{X}$

The ratio of the number of N and O atoms in $\mathbf{X}$ is: $(3.1\,\%/14.01) : (17.6\,\%/16.00) = 0.201 \approx 1 : 5 = 3 : 15$

CHECKPOINT 0.5 PTS

$\mathrm{N} : \mathrm{O} = 1 : 5$ in $\mathbf{X}$

The cation of $\mathbf{X}$ should be $[(\text{n-C}_4\text{H}_9)_4\text{N}]^+$; therefore, both Mo and O atoms are in the anion. Given the anion formula weight is 638.8,

there should be 4 Mo atoms and 15 O atoms in the anion;

The formula weight of the remaining part in the $\mathbf{X}$ anion is: $638.8 - 95.95 \times 4 - 16.00 \times 15 = 15.0$

Corresponding to 1 C and 3 H atoms;

Therefore, the composition of $\mathbf{X}$ is $[(\text{n-C}_4\text{H}_9)_4\text{N}]_3[\text{Mo}_4\text{O}_{15}\text{CH}_3]$

The carbon in the anion of $\mathbf{X}$ comes from formaldehyde hydrate $\mathrm{CH_2(OH)_2}$, so the chemical formula of $\mathbf{X}$ is: $[(\text{n-C}_4\text{H}_9)_4\text{N}]_3[\text{Mo}_4\text{O}_{12}(\text{CH}_2\text{O}_2)(\text{OH})]$

CHECKPOINT 2 PTS

The chemical formula of $\mathbf{X}$ is $[(\text{n-C}_4\text{H}_9)_4\text{N}]_3[\text{Mo}_4\text{O}_{12}(\text{CH}_2\text{O}_2)(\text{OH})]$

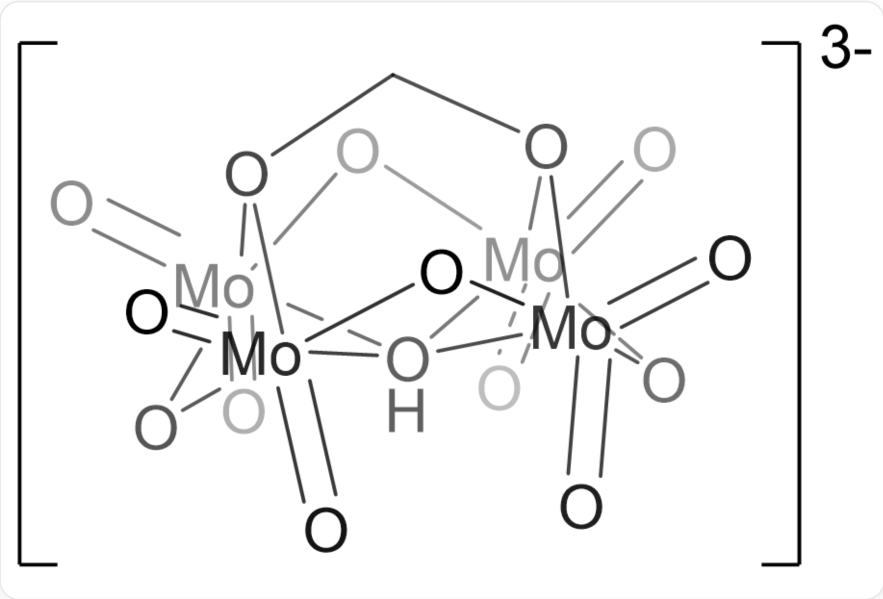
$\mathbf{X}$ contains $9 \times 4 \times 3 + 3 = 111$, statement 1 is incorrect.

The reaction of two molecules of $[(\text{n-C}_4\text{H}_9)_4\text{N}]_2\text{Mo}_2\text{O}_7$ with one molecule of $\mathrm{CH_2(OH)_2}$ will produce a strong base, statement 2 is incorrect: $2[(\text{n-C}_4\text{H}_9)_4\text{N}]_2\text{Mo}_2\text{O}_7 + \text{CH}_2(\text{OH})_2 \rightarrow [(\text{n-C}_4\text{H}_9)_4\text{N}]_3[\text{Mo}_4\text{O}_{12}(\text{CH}_2\text{O}_2)(\text{OH})] + [(\text{n-C}_4\text{H}_9)_4\text{N}]\text{OH}$

CHECKPOINT 1 PTS

The reaction will produce $[(\text{n-C}_4\text{H}_9)_4\text{N}]\text{OH}$

The chemical environment of Mo is the same, so $\mathrm{OH^-}$ must be in the middle of the $\mathrm{Mo_4}$ rectangle. There are 12 $\mathrm{O^{2-}}$, first consider 4 bridging and 4 terminal, at this time Mo is already four-coordinated, but in multinuclear cluster structures, it is generally six-coordinated. $\mathrm{CH_2O_2^{2-}}$ can be placed on the other side of $\mathrm{OH^-}$, and coordinated with 4 Mo at the same time, and the remaining 4 individual $\mathrm{O^{2-}}$ can be terminally coordinated with Mo. There are only 2 three-coordinated O, statement 3 is correct:



O=[Mo]1([OH]([Mo]2(O3)(O1)(=O)=O)([Mo]345(=O)=O)[Mo]67(O4)(=O)=O)(O6)(O2CO75)=O，带3个负电荷

CHECKPOINT 2 PTS

The anion structure of **X** is O=[Mo]1([OH]([Mo]2(O3)(O1)(=O)=O)([Mo]345(=O)=O)[Mo]67(O4)(=O)=O)(O6)(O2CO75)=O(with 3 negative charges)

Terminal oxygen O^{2-} has 2 negative charges and is a good σ donor and π donor; when there are more terminal O^{2-}, the Mo center is electron deficient and tends to form low-coordinated monomeric structures. Statement 4 is correct.

CHECKPOINT 1 PTS

Terminal oxygen O^{2-} has 2 negative charges and is a good σ donor and π donor; when there are more terminal O^{2-}, the Mo center is electron deficient and tends to form low-coordinated monomeric structures

Statements 3 and 4 are correct, choose K